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<https://github.com/hansmalvin/LastMile>

```
CREATE DATABASE performance_lab;
USE performance_lab;
CREATE TABLE employees (
  employee_id INT PRIMARY KEY AUTO_INCREMENT,
  first_name VARCHAR(50),
  last_name VARCHAR(50),
  email VARCHAR(100),
  department VARCHAR(50),
  salary DECIMAL(10, 2),
  hire_date DATE
);
DELIMITER //
CREATE PROCEDURE generate_employee_data(IN num_rows INT)
BEGIN
  DECLARE i INT DEFAULT 1;
  DECLARE first_name VARCHAR(50);
  DECLARE last_name VARCHAR(50);
  DECLARE email VARCHAR(100);
  DECLARE department VARCHAR(50);
  DECLARE salary DECIMAL(10, 2);
  DECLARE hire_date DATE;

  WHILE i <= num_rows DO
    SET first_name = CONCAT('FirstName', i);
    SET last_name = CONCAT('LastName', i);
    SET email = CONCAT('email', i, '@example.com');
    SET department = ELT(FLOOR(1 + RAND() * 5), 'Sales', 'Marketing', 'Engineering', 'HR', 'Finance');
    SET salary = FLOOR(50000 + RAND() * 50000);
    SET hire_date = DATE_SUB(CURDATE(), INTERVAL FLOOR(RAND() * 365 * 10) DAY);

    INSERT INTO employees (first_name, last_name, email, department, salary, hire_date)
    VALUES (first_name, last_name, email, department, salary, hire_date);

    SET i = i + 1;
  END WHILE;
END //

DELIMITER ;

CALL generate_employee_data(10000);
```

```
mysql> CREATE PROCEDURE generate_employee_data(IN num_rows INT)
-> BEGIN
-> DECLARE i INT DEFAULT 1;
-> DECLARE first_name VARCHAR(50);
-> DECLARE last_name VARCHAR(50);
-> DECLARE email VARCHAR(100);
-> DECLARE department VARCHAR(50);
-> DECLARE salary DECIMAL(10, 2);
-> DECLARE hire_date DATE;
-> WHILE i <= num_rows DO
-> SET first_name = CONCAT('FirstName', i);
-> SET last_name = CONCAT('LastName', i);
-> SET email = CONCAT('email', i, '@example.com');
-> SET department = ELT(FLOOR(1 + RAND() * 5), 'Sales', 'Marketing', 'Engineering', 'HR', 'Finance');
-> SET salary = FLOOR(50000 + RAND() * 50000);
-> SET hire_date = DATE_SUB(CURDATE(), INTERVAL FLOOR(RAND() * 3650) DAY);
-> INSERT INTO employees (first_name, last_name, email, department, salary, hire_date)
-> VALUES (first_name, last_name, email, department, salary, hire_date);
-> SET i = i + 1;
-> END WHILE;
-> END //
Query OK, 0 rows affected (0.037 sec)

mysql> DELIMITER ;
mysql> CALL generate_employee_data(10000);
Query OK, 1 row affected (2 min 39.627 sec)

mysql>
```

Step 1: Execute a Query Without an Index

SELECT * FROM employees WHERE department = 'Engineering';

9824	FirstName9824	LastName9824	email9824@example.com	Engineering	91895.00	2017-06-06
9827	FirstName9827	LastName9827	email9827@example.com	Engineering	99549.00	2023-04-21
9828	FirstName9828	LastName9828	email9828@example.com	Engineering	92037.00	2020-11-05
9839	FirstName9839	LastName9839	email9839@example.com	Engineering	91997.00	2020-10-12
9841	FirstName9841	LastName9841	email9841@example.com	Engineering	85033.00	2024-02-17
9846	FirstName9846	LastName9846	email9846@example.com	Engineering	52835.00	2018-10-05
9848	FirstName9848	LastName9848	email9848@example.com	Engineering	94224.00	2024-07-23
9852	FirstName9852	LastName9852	email9852@example.com	Engineering	88485.00	2023-10-10
9858	FirstName9858	LastName9858	email9858@example.com	Engineering	52776.00	2020-02-10
9860	FirstName9860	LastName9860	email9860@example.com	Engineering	67215.00	2022-07-04
9863	FirstName9863	LastName9863	email9863@example.com	Engineering	63481.00	2018-08-12
9866	FirstName9866	LastName9866	email9866@example.com	Engineering	93230.00	2017-08-23
9869	FirstName9869	LastName9869	email9869@example.com	Engineering	96148.00	2025-01-08
9873	FirstName9873	LastName9873	email9873@example.com	Engineering	99326.00	2020-03-25
9887	FirstName9887	LastName9887	email9887@example.com	Engineering	77362.00	2016-11-30
9899	FirstName9899	LastName9899	email9899@example.com	Engineering	53259.00	2026-04-29
9901	FirstName9901	LastName9901	email9901@example.com	Engineering	62364.00	2026-02-19
9908	FirstName9908	LastName9908	email9908@example.com	Engineering	90611.00	2018-06-30
9909	FirstName9909	LastName9909	email9909@example.com	Engineering	55323.00	2025-10-23
9912	FirstName9912	LastName9912	email9912@example.com	Engineering	57489.00	2021-07-23
9915	FirstName9915	LastName9915	email9915@example.com	Engineering	91291.00	2019-11-24
9926	FirstName9926	LastName9926	email9926@example.com	Engineering	99663.00	2021-11-21
9934	FirstName9934	LastName9934	email9934@example.com	Engineering	55959.00	2023-09-23
9937	FirstName9937	LastName9937	email9937@example.com	Engineering	99330.00	2020-12-24
9939	FirstName9939	LastName9939	email9939@example.com	Engineering	95173.00	2017-02-14
9953	FirstName9953	LastName9953	email9953@example.com	Engineering	51065.00	2018-11-08
9958	FirstName9958	LastName9958	email9958@example.com	Engineering	57582.00	2021-09-17
9960	FirstName9960	LastName9960	email9960@example.com	Engineering	85742.00	2018-04-29
9964	FirstName9964	LastName9964	email9964@example.com	Engineering	61505.00	2021-06-20
9974	FirstName9974	LastName9974	email9974@example.com	Engineering	70034.00	2021-03-15
9976	FirstName9976	LastName9976	email9976@example.com	Engineering	79577.00	2018-10-19
9978	FirstName9978	LastName9978	email9978@example.com	Engineering	95415.00	2023-08-03
9982	FirstName9982	LastName9982	email9982@example.com	Engineering	96138.00	2025-08-04
9987	FirstName9987	LastName9987	email9987@example.com	Engineering	70148.00	2018-04-04
9995	FirstName9995	LastName9995	email9995@example.com	Engineering	55407.00	2023-09-08

2034 rows in set (0.051 sec)

Step 2: Analyze the Query Execution Plan

EXPLAIN SELECT * FROM employees WHERE department = 'Engineering';

```
mysql> EXPLAIN SELECT * FROM employees WHERE department = 'Engineering';
+-----+-----+
| EXPLAIN |
+-----+-----+
| -> Filter: (employees.department = 'Engineering') (cost=1010 rows=986) |
| -> Table scan on employees (cost=1010 rows=9861) |
+-----+-----+
1 row in set (0.005 sec)

mysql>
```

Step 3: Create an Index on the department Column

CREATE INDEX idx_department ON employees (department);

```
mysql> CREATE INDEX idx_department ON employees (department);
Query OK, 0 rows affected (0.591 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
```

Step 4: Re-analyze the Query Execution Plan

```
EXPLAIN SELECT * FROM employees WHERE department = 'Engineering';
```

```
mysql> EXPLAIN SELECT * FROM employees WHERE department = 'Engineering';
+-----+-----+-----+-----+-----+-----+
| EXPLAIN                                                                                                                                                                |      |
+-----+-----+-----+-----+-----+-----+
| -> Index lookup on employees using idx_department (department = 'Engineering') (cost=276 rows=2034) |      |
|                                                                                                                                |      |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.009 sec)
```

```
mysql>
```

Step 5: Compare Query Execution Times

```
SELECT BENCHMARK(1000, (SELECT * FROM employees WHERE department = 'Engineering'));
```

```
mysql> SELECT BENCHMARK(1000, (SELECT count(*) FROM employees WHERE department = 'Engineering'));
+-----+
| BENCHMARK(1000, (SELECT count(*) FROM employees WHERE department = 'Engineering')) |
+-----+
| 0 |
+-----+
1 row in set (0.011 sec)
```

```
SELECT BENCHMARK(1000, (SELECT * FROM employees WHERE department = 'Engineering'));
```

```
mysql> SELECT BENCHMARK(1000, (SELECT count(*) FROM employees WHERE department = 'Engineering'));
+-----+
| BENCHMARK(1000, (SELECT count(*) FROM employees WHERE department = 'Engineering')) |
+-----+
| 0 |
+-----+
1 row in set (0.012 sec)
```

Step 6: Experiment with Different Queries

```
SELECT * FROM employees WHERE salary > 80000;
```

9864	FirstName9864	LastName9864	email9864@example.com	Sales	99225.00	2018-12-20
9866	FirstName9866	LastName9866	email9866@example.com	Engineering	93238.00	2017-08-23
9869	FirstName9869	LastName9869	email9869@example.com	Engineering	91388.00	2020-01-28
9872	FirstName9872	LastName9872	email9872@example.com	Engineering	99326.00	2020-03-25
9874	FirstName9874	LastName9874	email9874@example.com	Sales	83601.00	2025-10-19
9876	FirstName9876	LastName9876	email9876@example.com	Finance	96832.00	2017-03-27
9881	FirstName9881	LastName9881	email9881@example.com	HR	82213.00	2022-11-22
9882	FirstName9882	LastName9882	email9882@example.com	HR	97400.00	2022-11-13
9884	FirstName9884	LastName9884	email9884@example.com	Marketing	99573.00	2026-04-08
9886	FirstName9886	LastName9886	email9886@example.com	HR	91154.00	2018-03-22
9889	FirstName9889	LastName9889	email9889@example.com	Sales	84918.00	2023-06-21
9890	FirstName9890	LastName9890	email9890@example.com	Marketing	93799.00	2023-01-18
9891	FirstName9891	LastName9891	email9891@example.com	HR	95100.00	2023-01-14
9892	FirstName9892	LastName9892	email9892@example.com	Finance	81669.00	2022-06-13
9894	FirstName9894	LastName9894	email9894@example.com	Sales	91521.00	2018-10-30
9902	FirstName9902	LastName9902	email9902@example.com	Marketing	87819.00	2019-06-15
9903	FirstName9903	LastName9903	email9903@example.com	Sales	91669.00	2020-02-07
9908	FirstName9908	LastName9908	email9908@example.com	Engineering	90611.00	2018-06-30
9914	FirstName9914	LastName9914	email9914@example.com	Marketing	93858.00	2024-03-12
9915	FirstName9915	LastName9915	email9915@example.com	Engineering	91291.00	2019-11-24
9916	FirstName9916	LastName9916	email9916@example.com	HR	90121.00	2021-02-21
9917	FirstName9917	LastName9917	email9917@example.com	HR	95211.00	2021-05-24
9919	FirstName9919	LastName9919	email9919@example.com	Sales	93705.00	2023-03-31
9920	FirstName9920	LastName9920	email9920@example.com	Finance	85920.00	2018-05-02
9921	FirstName9921	LastName9921	email9921@example.com	Finance	93570.00	2018-06-14
9923	FirstName9923	LastName9923	email9923@example.com	Finance	89281.00	2024-01-14
9926	FirstName9926	LastName9926	email9926@example.com	Engineering	99663.00	2021-11-21
9927	FirstName9927	LastName9927	email9927@example.com	Marketing	95815.00	2018-01-25
9935	FirstName9935	LastName9935	email9935@example.com	Finance	99846.00	2025-11-28
9937	FirstName9937	LastName9937	email9937@example.com	Engineering	99040.00	2020-08-29
9939	FirstName9939	LastName9939	email9939@example.com	Engineering	95171.00	2017-02-14
9940	FirstName9940	LastName9940	email9940@example.com	Finance	87858.00	2025-08-26
9942	FirstName9942	LastName9942	email9942@example.com	HR	94782.00	2024-12-10
9943	FirstName9943	LastName9943	email9943@example.com	Sales	84252.00	2022-09-28
9944	FirstName9944	LastName9944	email9944@example.com	HR	83256.00	2025-07-31
9945	FirstName9945	LastName9945	email9945@example.com	Marketing	85658.00	2022-05-10
9949	FirstName9949	LastName9949	email9949@example.com	Marketing	83667.00	2022-09-01
9951	FirstName9951	LastName9951	email9951@example.com	HR	96540.00	2021-03-05
9955	FirstName9955	LastName9955	email9955@example.com	HR	91320.00	2020-02-21
9959	FirstName9959	LastName9959	email9959@example.com	Finance	96756.00	2025-07-23
9960	FirstName9960	LastName9960	email9960@example.com	Engineering	85742.00	2018-04-29
9961	FirstName9961	LastName9961	email9961@example.com	Finance	97819.00	2025-06-04
9962	FirstName9962	LastName9962	email9962@example.com	HR	96496.00	2019-02-07
9967	FirstName9967	LastName9967	email9967@example.com	Finance	96612.00	2019-01-02
9969	FirstName9969	LastName9969	email9969@example.com	Finance	86260.00	2017-10-11
9970	FirstName9970	LastName9970	email9970@example.com	Sales	99482.00	2020-03-30
9972	FirstName9972	LastName9972	email9972@example.com	HR	80742.00	2017-12-22
9977	FirstName9977	LastName9977	email9977@example.com	Sales	87925.00	2010-01-07
9978	FirstName9978	LastName9978	email9978@example.com	Engineering	95101.00	2023-08-03
9980	FirstName9980	LastName9980	email9980@example.com	Sales	96904.00	2024-08-13
9981	FirstName9981	LastName9981	email99			

EXPLAIN SELECT * FROM employees WHERE salary > 80000;

```
mysql> EXPLAIN SELECT * FROM employees WHERE salary > 80000;
+-----+
| EXPLAIN |
+-----+
| -> Filter: (employees.salary > 80000.00) (cost=1010 rows=3287) |
| -> Table scan on employees (cost=1010 rows=9861) |
| |
+-----+
1 row in set (0.006 sec)
```

CREATE INDEX idx_salary ON employees (salary);

```
mysql> CREATE INDEX idx_salary ON employees (salary);
Query OK, 0 rows affected (0.477 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

SELECT * FROM employees WHERE salary > 80000;

```
9942 | FirstName9942 | LastName9942 | email9942@example.com | HR | 84782.00 | 2024-12-10 |
9943 | FirstName9943 | LastName9943 | email9943@example.com | Sales | 84252.00 | 2022-09-28 |
9944 | FirstName9944 | LastName9944 | email9944@example.com | HR | 83256.00 | 2025-07-31 |
9945 | FirstName9945 | LastName9945 | email9945@example.com | Marketing | 85658.00 | 2022-05-10 |
9949 | FirstName9949 | LastName9949 | email9949@example.com | Marketing | 83667.00 | 2022-09-01 |
9951 | FirstName9951 | LastName9951 | email9951@example.com | HR | 96540.00 | 2021-03-05 |
9955 | FirstName9955 | LastName9955 | email9955@example.com | HR | 81939.00 | 2023-03-22 |
9959 | FirstName9959 | LastName9959 | email9959@example.com | Finance | 96756.00 | 2025-07-23 |
9960 | FirstName9960 | LastName9960 | email9960@example.com | Engineering | 85742.00 | 2018-04-29 |
9961 | FirstName9961 | LastName9961 | email9961@example.com | Finance | 97019.00 | 2025-06-04 |
9962 | FirstName9962 | LastName9962 | email9962@example.com | HR | 96496.00 | 2019-02-07 |
9967 | FirstName9967 | LastName9967 | email9967@example.com | Finance | 96612.00 | 2019-01-02 |
9969 | FirstName9969 | LastName9969 | email9969@example.com | Finance | 86260.00 | 2017-10-11 |
9970 | FirstName9970 | LastName9970 | email9970@example.com | Sales | 99482.00 | 2020-03-30 |
9972 | FirstName9972 | LastName9972 | email9972@example.com | HR | 80742.00 | 2017-12-22 |
9977 | FirstName9977 | LastName9977 | email9977@example.com | Sales | 87425.00 | 2019-01-07 |
9978 | FirstName9978 | LastName9978 | email9978@example.com | Engineering | 95415.00 | 2023-08-03 |
9980 | FirstName9980 | LastName9980 | email9980@example.com | Sales | 96904.00 | 2024-08-13 |
9981 | FirstName9981 | LastName9981 | email9981@example.com | Sales | 87031.00 | 2020-11-18 |
9982 | FirstName9982 | LastName9982 | email9982@example.com | Engineering | 96138.00 | 2025-08-04 |
9983 | FirstName9983 | LastName9983 | email9983@example.com | HR | 91195.00 | 2023-06-18 |
9984 | FirstName9984 | LastName9984 | email9984@example.com | Finance | 99902.00 | 2025-08-20 |
9986 | FirstName9986 | LastName9986 | email9986@example.com | HR | 93463.00 | 2024-05-01 |
9988 | FirstName9988 | LastName9988 | email9988@example.com | Finance | 88339.00 | 2023-03-09 |
9991 | FirstName9991 | LastName9991 | email9991@example.com | Sales | 85364.00 | 2021-08-24 |
9992 | FirstName9992 | LastName9992 | email9992@example.com | Marketing | 86983.00 | 2026-04-09 |
10000 | FirstName10000 | LastName10000 | email10000@example.com | Finance | 91247.00 | 2019-07-27 |
+-----+
4015 rows in set (0.057 sec)
```

EXPLAIN SELECT * FROM employees WHERE salary > 80000;

```
mysql> EXPLAIN SELECT * FROM employees WHERE salary > 80000;
+-----+
| EXPLAIN |
+-----+
| -> Filter: (employees.salary > 80000.00) (cost=1010 rows=4015) |
| -> Table scan on employees (cost=1010 rows=9861) |
| |
+-----+
1 row in set (0.005 sec)
```

SELECT * FROM employees WHERE hire_date > '2020-01-01';

```

9935 | FirstName9935 | LastName9935 | email9935@example.com | Finance | 99046.00 | 2025-11-28 |
9936 | FirstName9936 | LastName9936 | email9936@example.com | Marketing | 62794.00 | 2021-11-19 |
9937 | FirstName9937 | LastName9937 | email9937@example.com | Engineering | 99330.00 | 2020-12-24 |
9940 | FirstName9940 | LastName9940 | email9940@example.com | Finance | 87858.00 | 2025-08-26 |
9942 | FirstName9942 | LastName9942 | email9942@example.com | HR | 94702.00 | 2024-12-10 |
9943 | FirstName9943 | LastName9943 | email9943@example.com | Sales | 84252.00 | 2022-09-28 |
9944 | FirstName9944 | LastName9944 | email9944@example.com | HR | 83256.00 | 2025-07-31 |
9945 | FirstName9945 | LastName9945 | email9945@example.com | Marketing | 85658.00 | 2022-05-10 |
9947 | FirstName9947 | LastName9947 | email9947@example.com | Marketing | 71269.00 | 2021-11-20 |
9948 | FirstName9948 | LastName9948 | email9948@example.com | Finance | 71974.00 | 2023-01-18 |
9949 | FirstName9949 | LastName9949 | email9949@example.com | Marketing | 83667.00 | 2022-09-01 |
9950 | FirstName9950 | LastName9950 | email9950@example.com | Finance | 50069.00 | 2020-12-02 |
9951 | FirstName9951 | LastName9951 | email9951@example.com | HR | 96540.00 | 2021-03-05 |
9955 | FirstName9955 | LastName9955 | email9955@example.com | HR | 81939.00 | 2023-03-22 |
9956 | FirstName9956 | LastName9956 | email9956@example.com | HR | 65274.00 | 2020-07-18 |
9957 | FirstName9957 | LastName9957 | email9957@example.com | Finance | 59625.00 | 2026-05-01 |
9958 | FirstName9958 | LastName9958 | email9958@example.com | Engineering | 57582.00 | 2021-09-17 |
9959 | FirstName9959 | LastName9959 | email9959@example.com | Finance | 96756.00 | 2025-07-23 |
9961 | FirstName9961 | LastName9961 | email9961@example.com | Finance | 97019.00 | 2025-06-04 |
9963 | FirstName9963 | LastName9963 | email9963@example.com | Finance | 50211.00 | 2021-03-21 |
9964 | FirstName9964 | LastName9964 | email9964@example.com | Engineering | 61505.00 | 2021-06-20 |
9965 | FirstName9965 | LastName9965 | email9965@example.com | HR | 64702.00 | 2024-03-15 |
9968 | FirstName9968 | LastName9968 | email9968@example.com | Finance | 59310.00 | 2023-05-22 |
9970 | FirstName9970 | LastName9970 | email9970@example.com | Sales | 99482.00 | 2020-03-30 |
9972 | FirstName9972 | LastName9972 | email9972@example.com | Marketing | 60626.00 | 2026-01-30 |
9974 | FirstName9974 | LastName9974 | email9974@example.com | Engineering | 70034.00 | 2021-03-15 |
9975 | FirstName9975 | LastName9975 | email9975@example.com | Marketing | 65881.00 | 2021-08-23 |
9978 | FirstName9978 | LastName9978 | email9978@example.com | Engineering | 95415.00 | 2023-08-03 |
9979 | FirstName9979 | LastName9979 | email9979@example.com | HR | 72789.00 | 2023-04-09 |
9980 | FirstName9980 | LastName9980 | email9980@example.com | Sales | 96904.00 | 2024-08-13 |
9981 | FirstName9981 | LastName9981 | email9981@example.com | Sales | 87031.00 | 2020-11-18 |
9982 | FirstName9982 | LastName9982 | email9982@example.com | Engineering | 96138.00 | 2025-08-04 |
9983 | FirstName9983 | LastName9983 | email9983@example.com | HR | 91195.00 | 2023-06-18 |
9984 | FirstName9984 | LastName9984 | email9984@example.com | Finance | 99902.00 | 2025-08-20 |
9986 | FirstName9986 | LastName9986 | email9986@example.com | HR | 93463.00 | 2024-05-01 |
9988 | FirstName9988 | LastName9988 | email9988@example.com | Finance | 88339.00 | 2023-03-09 |
9989 | FirstName9989 | LastName9989 | email9989@example.com | Marketing | 73144.00 | 2021-09-06 |
9991 | FirstName9991 | LastName9991 | email9991@example.com | Sales | 85364.00 | 2021-08-24 |
9992 | FirstName9992 | LastName9992 | email9992@example.com | Marketing | 86983.00 | 2026-04-09 |
9994 | FirstName9994 | LastName9994 | email9994@example.com | Marketing | 54470.00 | 2022-12-19 |
9995 | FirstName9995 | LastName9995 | email9995@example.com | Engineering | 55407.00 | 2023-09-08 |
9996 | FirstName9996 | LastName9996 | email9996@example.com | Sales | 61884.00 | 2024-09-07 |
9997 | FirstName9997 | LastName9997 | email9997@example.com | Sales | 54735.00 | 2025-03-05 |
9999 | FirstName9999 | LastName9999 | email9999@example.com | Finance | 60081.00 | 2025-07-05 |
+-----+-----+-----+-----+-----+-----+-----+
6356 rows in set (0.065 sec)

```

EXPLAIN SELECT * FROM employees WHERE hire_date > '2020-01-01';

```

mysql> EXPLAIN SELECT * FROM employees WHERE hire_date > '2020-01-01';
+-----+-----+-----+-----+-----+-----+-----+
| EXPLAIN |
+-----+-----+-----+-----+-----+-----+
| -> Filter: (employees.hire_date > DATE'2020-01-01') (cost=1010 rows=3287) |
| -> Table scan on employees (cost=1010 rows=9861) |
| |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.005 sec)

```

CREATE INDEX idx_hire_date ON employees (hire_date);

```

mysql> CREATE INDEX idx_hire_date ON employees (hire_date);
Query OK, 0 rows affected (0.570 sec)
Records: 0 Duplicates: 0 Warnings: 0

```

SELECT * FROM employees WHERE hire_date > '2020-01-01';

```

9940 | FirstName9940 | LastName9940 | email9940@example.com | Finance | 87858.00 | 2025-08-26 |
9942 | FirstName9942 | LastName9942 | email9942@example.com | HR | 94702.00 | 2024-12-10 |
9943 | FirstName9943 | LastName9943 | email9943@example.com | Sales | 84252.00 | 2022-09-28 |
9944 | FirstName9944 | LastName9944 | email9944@example.com | HR | 83256.00 | 2025-07-31 |
9945 | FirstName9945 | LastName9945 | email9945@example.com | Marketing | 85658.00 | 2022-05-10 |
9947 | FirstName9947 | LastName9947 | email9947@example.com | Marketing | 71269.00 | 2021-11-20 |
9948 | FirstName9948 | LastName9948 | email9948@example.com | Finance | 71974.00 | 2023-01-18 |
9949 | FirstName9949 | LastName9949 | email9949@example.com | Marketing | 83667.00 | 2022-09-01 |
9950 | FirstName9950 | LastName9950 | email9950@example.com | Finance | 50069.00 | 2020-12-02 |
9951 | FirstName9951 | LastName9951 | email9951@example.com | HR | 96540.00 | 2021-03-05 |
9955 | FirstName9955 | LastName9955 | email9955@example.com | HR | 81939.00 | 2023-03-22 |
9956 | FirstName9956 | LastName9956 | email9956@example.com | HR | 65274.00 | 2020-07-18 |
9957 | FirstName9957 | LastName9957 | email9957@example.com | Finance | 59625.00 | 2026-05-01 |
9958 | FirstName9958 | LastName9958 | email9958@example.com | Engineering | 57582.00 | 2021-09-17 |
9959 | FirstName9959 | LastName9959 | email9959@example.com | Finance | 96756.00 | 2025-07-23 |
9961 | FirstName9961 | LastName9961 | email9961@example.com | Finance | 97019.00 | 2025-06-04 |
9963 | FirstName9963 | LastName9963 | email9963@example.com | Finance | 50211.00 | 2021-03-21 |
9964 | FirstName9964 | LastName9964 | email9964@example.com | Engineering | 61505.00 | 2021-06-20 |
9965 | FirstName9965 | LastName9965 | email9965@example.com | HR | 64702.00 | 2024-03-15 |
9968 | FirstName9968 | LastName9968 | email9968@example.com | Finance | 59310.00 | 2023-05-22 |
9970 | FirstName9970 | LastName9970 | email9970@example.com | Sales | 99482.00 | 2020-03-30 |
9973 | FirstName9973 | LastName9973 | email9973@example.com | Marketing | 60626.00 | 2026-01-30 |
9974 | FirstName9974 | LastName9974 | email9974@example.com | Engineering | 70034.00 | 2021-03-15 |
9975 | FirstName9975 | LastName9975 | email9975@example.com | Marketing | 65881.00 | 2021-08-23 |
9978 | FirstName9978 | LastName9978 | email9978@example.com | Engineering | 95415.00 | 2023-08-03 |
9979 | FirstName9979 | LastName9979 | email9979@example.com | HR | 72789.00 | 2023-04-09 |
9980 | FirstName9980 | LastName9980 | email9980@example.com | Sales | 96904.00 | 2024-08-13 |
9981 | FirstName9981 | LastName9981 | email9981@example.com | Sales | 87031.00 | 2020-11-18 |
9982 | FirstName9982 | LastName9982 | email9982@example.com | Engineering | 96138.00 | 2025-08-04 |
9983 | FirstName9983 | LastName9983 | email9983@example.com | HR | 91195.00 | 2023-06-18 |
9984 | FirstName9984 | LastName9984 | email9984@example.com | Finance | 99902.00 | 2025-08-20 |
9986 | FirstName9986 | LastName9986 | email9986@example.com | HR | 93463.00 | 2024-05-01 |
9988 | FirstName9988 | LastName9988 | email9988@example.com | Finance | 88339.00 | 2023-03-09 |
9989 | FirstName9989 | LastName9989 | email9989@example.com | Marketing | 73144.00 | 2021-09-06 |
9991 | FirstName9991 | LastName9991 | email9991@example.com | Sales | 85364.00 | 2021-08-24 |
9992 | FirstName9992 | LastName9992 | email9992@example.com | Marketing | 86983.00 | 2026-04-09 |
9994 | FirstName9994 | LastName9994 | email9994@example.com | Marketing | 54470.00 | 2022-12-19 |
9995 | FirstName9995 | LastName9995 | email9995@example.com | Engineering | 55407.00 | 2023-09-08 |
9996 | FirstName9996 | LastName9996 | email9996@example.com | Sales | 61884.00 | 2024-09-07 |
9997 | FirstName9997 | LastName9997 | email9997@example.com | Sales | 54735.00 | 2025-03-05 |
9999 | FirstName9999 | LastName9999 | email9999@example.com | Finance | 60081.00 | 2025-07-05 |
+-----+-----+-----+-----+-----+-----+-----+
6356 rows in set (0.067 sec)

```

EXPLAIN SELECT * FROM employees WHERE hire_date > '2020-01-01';

```

mysql> EXPLAIN SELECT * FROM employees WHERE hire_date > '2020-01-01';
+-----+-----+-----+-----+-----+-----+-----+
| EXPLAIN |
+-----+-----+-----+-----+-----+-----+
| -> Filter: (employees.hire_date > DATE'2020-01-01') (cost=1010 rows=6356) |
|   -> Table scan on employees (cost=1010 rows=9861) |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.005 sec)

```

Summary

In this lab, we evaluated MySQL query efficiency by comparing performance before and after indexing. Without indexes, the database was forced to perform a full table scan (type: ALL), inspecting every record in the 10,000-row dataset, which resulted in significant latency. Once indexes were applied to specific columns like department, the system transitioned to index scans. This shift drastically reduced the volume of data examined and accelerated execution times. These performance gains were further validated by the BENCHMARK() function and remained consistent when testing other indexed fields, such as salary and hire_date.

Conclusions

Indexes are vital for optimizing database performance, as they allow for rapid data retrieval without the need for exhaustive table scans. However, they involve specific trade-offs: they occupy additional storage space and can introduce overhead that slows down write-heavy operations like INSERT, UPDATE, and DELETE. Consequently, an effective indexing strategy requires a careful balance between query speed and the resource costs associated with maintaining them.